

Program : Diploma in Computer Application and Business Management	
Course Code : 2259	Course Title: Basics of Computer Hardware & Networking
Semester : 2	Credits: 1
Course Category: Foundation Course	
Periods per week: 2 (L:0, T:0, P:2)	Periods per semester: 30

Course Objectives:

- To provide a hands-on experience in various hardware components of computer system.
- To be able to install different operating systems and demonstrate trouble shooting & networking in computer systems.

Course Prerequisites: Nil

Course Outcomes:

On completion of the course, students will be able to:

CO_n	Description	Duration (Hours)	Cognitive Level
CO1	Identify various hardware components of the computer system	6	Applying
CO2	Assembling various hardware components	6	Applying
CO3	Installation of OS, application software and similar other components	9	Applying
CO4	Troubleshooting and Networking	6	Applying
	Lab Exam	3	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3		3	3			
CO3	3			3			
CO4	3	3		3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Name of the Experiment	Duration (Hours)	Cognitive Level
CO1	Identify various hardware components of the computer system		
M1.1	Introduction to Fundamentals of Computers - Types of Computers-Machine-assembly and High-level languages- - Data Organization-Types of memory-Secondary storage Devices-I/O devices	2	Applying
M1.2	Demonstration of mother board – identification of processor sockets and CPU support, power connector, keyboard connector, BIOS, bus slots, chipset, cache memory, co-processors, CMOS battery, etc	2	Applying
M1.3	Demonstration of SMPS and its connections, Demonstration of HDDs and various monitors (CRT, TFT, LCD)	2	Applying
CO2	Assembling various hardware components.		
M2.1	Demonstrating the system assembly – CPU – cooler fan, SMPs – memory monitor – keyboard – HDD	3	Applying
M2.2	Identifying the parts of laptop computer- its assembling and disassembling	3	Applying
	Lab Exam – I	1½	
CO3	Installation of OS, application software and similar other components.		
M3.1	Installation windows and Linux operating system from CDs and HDDs – installation of drivers of display	3	Applying

M3.2	Demonstration of CDs – Drives – Drive installation procedure – CD writer installation and writing practice	3	Applying
M3.3	Installation of Various Application Software	3	Applying
CO4	Troubleshooting and Networking.		
M4.1	Various troubles and rectification procedures of motherboard, CPU, etc. - OS related problems (WINDOWS &, LINUX). Antivirus software installation	2	Applying
M 4.2	Demonstration of Network Components – Switches – Routers Internet Modems – UTP Cable Crimping - printer connection – basic troubleshooting on networks	1	Applying
M4.3	Internet – Intranet - E-mail - Internet Sharing – Network Printers – Trouble Shooting	1	Applying
M4.4	Open ended experiments	2	
	Lab Exam – II	1½	

Text / Reference:

T/R	Book Title/Author
R1	Vikas Gupta, Comdex Hardware & Networking Course Kit , Revised and Upgraded, BSC publishing
R2	Networking A Beginner's Guide Sixth Edition by Hallberg Bruce, Published by McGraw Hill Education India
R3	Stephen J. Bigelow, Trouble shooting maintaining & Repairing of PCs , 5th ed. Tata McGraw-Hill, 2001

Online Resources:

Sl.No	Website Link
1	https://en.wikibooks.org/wiki/How_To_Assemble_A_Desktop_PC/Assembly
2	https://itfundamentals.in
3	www.laptopcareiit.com/crash-course-in-computer-hardware.html
4	https://www.coursera.org/lecture/technical...fundamentals/basics-of-networking-p1kFa
5	https://en.wikiversity.org/wiki/IT_Fundamentals/Hardware